A. Per-head update step

k_h v_h q_h d_h g_h e_h w_h

$\downarrow$ L^2 norm q, k $\downarrow$

$$r_h = S_h^T (e_h \cdot k_h) \quad \text{(gated read at address } e_h \cdot k_h\text{)}$$

$\downarrow$

$$\delta_h = w_h \cdot v_h - r_h \quad \text{(prediction error)}$$

$\downarrow$

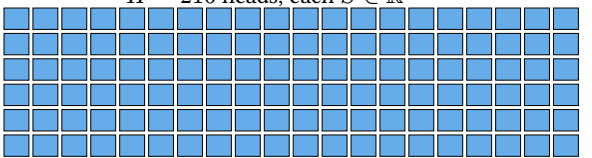
$$S_h \leftarrow \tanh\left(d_h \cdot S_h + k_h \delta_h^T\right)$$

$\downarrow$

$$y_h = \text{silu}(g_h) \odot S_h^T q_h$$

B. Multi-programmed shape (per layer, per batch element)

$H = 216$ heads, each $S \in \mathbb{R}^{32 \times 32}$



⋯ 216 small recurrent programs per layer ⋯

At batch size 4 and depth 11:

9,504 independent programs per token

(the E88 predecessor's 22,200-at-batch-5 count is Lean-certified: `emender_1p27B_programs_per_batch_token_bs5`*)*

Each program: a 32×32 state tile in registers/SRAM; serial along time; parallel across $H \times$ batch.